Renewal Theory

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References

- S.M. Ross, “Stochastic Processes”: Chapter 3
- R. Wolff: Chapter 2 and Appendix A-1
- S.M. Ross, “Introduction to Probability Models”: Chapter 7

Motivation

- Recall: interarrival times for Poisson process:
  - i.i.d. exponentially distributed r.v.s
- Generalization:
  - counting process for which interarrival times are i.i.d. with arbitrary CDF $F(\cdot)$
  - called "*renewal process*"
- E.g.:
  - Suppose we have infinite supply of lightbulbs whose lifetimes are i.i.d.
  - We use single lightbulb at a time
  - When it fails, we immediately replace it with a new one
  - $\{N(t), t \geq 0\}$ is a renewal process, where $N(t)$: number of lightbulbs that have failed until time t

Renewal Process

- Let $\{X_n : n = 1,2, \dots\}$ be sequence of non-negative i.i.d. r.v.s with CDF $F(\cdot)$
- X_n: time between $(n-1)$st and nth events
- To avoid trivialities, assume that:
  - $F(0) = P\{X_n = 0\} < 1$
- Let $\mu = E[X_n]$
- Note that $0 < \mu \leq \infty$
- Let:
  - $S_0 = 0$ and $S_n = \sum_{i=1}^{n} X_i$ for $n \in \{1,2, \dots\}$
- S_n: time of nth event

Renewal Process (contd.)

- Recall:
  - S_n: time of nth event
- Let $N(t)$: number of events that occur up to time t
- The counting process $\{N(t), t \geq 0\}$ is called a *renewal process*
- $N(t)$ in terms of S_n:
  - $\max\{n: S_n \leq t\}$
- **Proposition**: For each $t \geq 0$, $N(t)$ is finite w.p. 1
- **Proof**:
  - By SLLN, $\frac{S_n}{n} \to \mu$ a.s. as $n \to \infty$
  - Since $\mu > 0$, $S_n \to \infty$ w.p. 1 as $n \to \infty$
- Hence, an infinite number of renewals cannot occur in a finite amount of time

Distribution of $N(t)$

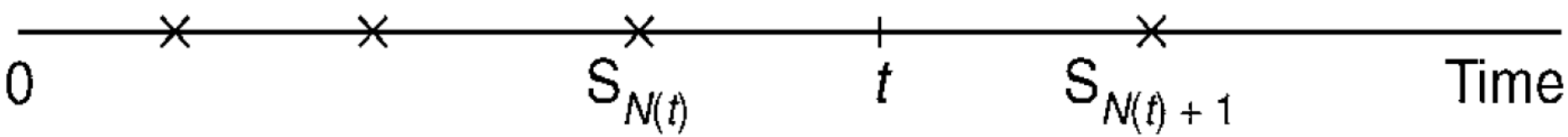
- Recall:
  - ❑CDF of X_n is $F(\cdot)$
  - ❑ $S_n = \sum_{i=1}^{n} X_i$
- For $n \geq 2$, PDF or PMF of S_n:
  - ❑convolution of PDF or PMF corresponding to $F(\cdot)$ with itself $(n-1)$ times
- Denote CDF of S_n by $F_n(\cdot)$
- PMF $P(N(t) = n)$:
  - ❑$P(N(t) \geq n) - P(N(t) \geq n+1)$
  - ❑$P(S_n \leq t) - P(S_{n+1} \leq t)$
  - ❑$F_n(t) - F_{n+1}(t)$

Renewal Function

- Recall:
  - $N(t)$: number of events that occur up to time t
- Let $m(t) = E[N(t)]$:
  - called "*renewal function*"
- **Proposition**: For all $t \geq 0$, $m(t) = \sum_{n=1}^{\infty} F_n(t)$
- **Proof**:
  - $N(t) = \sum_{n=1}^{\infty} 1_{\{S_n \leq t\}}$
  - Take expectations on both sides
- Recall:
  - For each $t \geq 0$, $N(t)$ is finite w.p. 1
- **Proposition**: For each $t \geq 0$, $m(t) < \infty$
- Proof omitted

Long-Run Renewal Rate

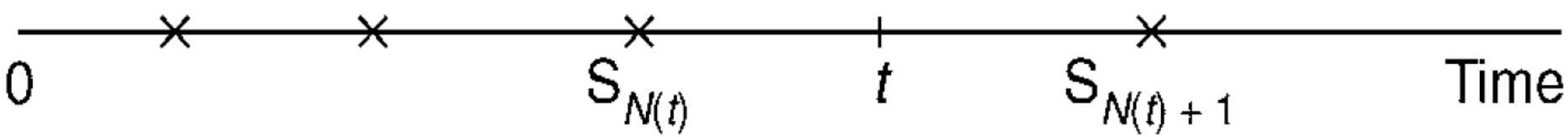
- Let $N(\infty) = \lim_{t \to \infty} N(t)$:
  - ❑ denotes total number of renewals that occur
- **Proposition**: $N(\infty) = \infty$ w.p. 1
- **Proof**:
  - ❑ $P\{N(\infty) < \infty\}$:
    - ○ $= P\{X_n = \infty \text{ for some } n\}$
    - ○ $= P\{\bigcup_{n=1}^{\infty}\{X_n = \infty\}\}$
    - ○ $\leq \sum_{n=1}^{\infty} P(X_n = \infty)$
    - ○ $= 0$
- Want to know rate at which $N(t)$ goes to ∞ as $t \to \infty$
- Note that:
  - ❑ $S_{N(t)}$: time of last renewal before or at time t
  - ❑ $S_{N(t)+1}$: time of first renewal after time t



Ref: S.M. Ross, "Stochastic Processes", 2nd ed.

Long-Run Renewal Rate (contd.)

- **Proposition**: W.p. 1, $\frac{N(t)}{t} \to \frac{1}{\mu}$ as $t \to \infty$
- **Proof**:
  - ❑ $S_{N(t)} \leq t < S_{N(t)+1}$
  - ❑ So, $\frac{S_{N(t)}}{N(t)} \leq \frac{t}{N(t)} < \left(\frac{S_{N(t)+1}}{N(t)+1}\right)\left(\frac{N(t)+1}{N(t)}\right)$
  - ❑ Take limits as $t \to \infty$ and use SLLN
- Hence, long-run renewal rate equals $\frac{1}{\mu}$
- So, $\frac{1}{\mu}$ called *rate* of renewal process



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Example 1

- Beverly has a radio that works on a single battery
- When battery in use fails, she immediately replaces it with new battery
- Lifetime of battery:
  - Uniform$[30,60]$
- Rate at which Beverly needs to change batteries:
  - $\frac{1}{45}$
- Now, instead suppose Beverly does not keep any spare batteries on hand
- Each time failure occurs, she goes and buys new battery
- Amount of time it takes her to get new battery:
  - Uniform$[0,1]$
- Rate at which Beverly needs to change batteries:
  - $\frac{1}{45.5}$

Example 2

- Potential customers arrive at a single-server bank according to a Poisson process with rate λ
  - enters the bank only if the server is free when he/ she arrives, else goes home
- Amount of time spent in bank by entering customer:
  - r.v. with CDF $G(\cdot)$ and mean μ_G
- Rate at which customers enter the bank:
  - $\frac{1}{\mu_G+\frac{1}{\lambda}}$
- Proportion of potential customers who actually enter the bank:
  - $\frac{1}{1+\lambda\mu_G}$
- E.g.:
  - Suppose $\lambda = \mu_G = 2$
  - Then only one customer out of five actually enters the bank

Elementary Renewal Theorem

- Recall:
  1) **Proposition**: W.p. 1, $\frac{N(t)}{t} \to \frac{1}{\mu}$ as $t \to \infty$
  - ❑ $m(t) = E[N(t)]$
- **Theorem** (Elementary Renewal Theorem):
  2) $\frac{m(t)}{t} \to \frac{1}{\mu}$ as $t \to \infty$
- At first glance, seems that 2) easily follows from 1):
  - ❑ not true
- E.g.:
  - ❑ Let U: Uniform$[0,1]$
  - ❑ For $n \geq 1$, let:
    - $Y_n = \begin{cases} 0, & \text{if } U > \frac{1}{n}, \\ n, & \text{else.} \end{cases}$
  - ❑ So, for all sufficiently large n, $Y_n = 0$
  - ❑ Hence, w.p. 1, $Y_n \to 0$ as $n \to \infty$
  - ❑ However, $E[Y_n]$:
    - 1 for all n
- To prove 2), we need concept of "*stopping times*" and *Wald's equation*

Stopping Time

- Let $X_1, X_2, \ldots$ be sequence of independent r.v.s
- An integer-valued r.v. N is said to be a stopping time for this sequence if:
  - For all $n \in \{1,2, \ldots\}$, the event $\{N = n\}$ is independent of $X_{n+1}, X_{n+2}, \ldots$
- *Intuition*:
  - We observe the r.v.s X_n in sequential order
  - N denotes the number observed before stopping
  - Decision of whether to stop at time n must be based only on information present at time n, not on any future information

Example 1

- Let X_n, $n \in \{1,2,\dots\}$ be independent and s.t.:
  - $P(X_n = 0) = P(X_n = 1) = \frac{1}{2}$ for all $n \in \{1,2,\dots\}$
- Let $N = \min\{n: X_1 + \cdots + X_n = 10\}$
- Is N a stopping time?
  - Yes
- E.g.: N may be stopping time of experiment that repeatedly flips a fair coin and stops when number of heads reaches 10

Example 2

- Let X_n, $n \in \{1,2,\dots\}$ be independent and s.t.:
  - $P(X_n = -1) = P(X_n = 1) = \frac{1}{2}$ for all $n \in \{1,2,\dots\}$
- Let $N = \min\{n: X_1 + \cdots + X_n = 1\}$
- Is N a stopping time?
  - Yes
- E.g.: N may be stopping time for gambler who:
  - on each play is equally likely to either win or lose 1 unit
  - stops the first time he/ she is ahead

Example 3

- Consider a sequence of i.i.d. coin tosses with $P(\text{heads}) = p \in (0,1)$
- For $n \in \{1,2,\dots\}$:
  - $X_n = \begin{cases} 1, & \text{if heads occurs on toss } n, \\ 0, & \text{else.} \end{cases}$
- Let:
  - N be toss on which first heads occurs
  - $K = N - 1$ be number of tails prior to first heads
- Which of N and K is/ are stopping time(s)?
  - N is a stopping time, K is not

Wald's Equation

- **Theorem**:
  - Suppose:
    - $X_1, X_2, \ldots$ is a sequence of i.i.d. r.v.s such that for all $n \geq 1$, $E[|X_n|] = E[|X|] < \infty$
    - N is a stopping time for this sequence s.t. $E[N] < \infty$
  - Then:
    - $E[\sum_{n=1}^{N} X_n] = E[N]E[X]$
- **Proof**:
  - $S_N = \sum_{n=1}^{N} X_n$:
    - $\sum_{n=1}^{\infty} X_n \, 1_{\{N \geq n\}}$
  - So, $E[S_N]$:
    - $\sum_{n=1}^{\infty} E\left[X_n 1_{\{N \geq n\}}\right]$
    - $\sum_{n=1}^{\infty} E[X_n] \, E\left[1_{\{N \geq n\}}\right]$ since the event $\{N \geq n\}$ is independent of X_n
    - $E[X](\sum_{n=1}^{\infty} P\{N \geq n\})$
    - $E[X] \, E[N]$

Example 1 (contd.)

- Recall:
  - Let X_n, $n \in \{1,2,\dots\}$ be independent and s.t.:
    - $P(X_n = 0) = P(X_n = 1) = \frac{1}{2}$ for all $n \in \{1,2,\dots\}$
  - Let $N = \min\{n: X_1 + \cdots + X_n = 10\}$
  - N is a stopping time
- $E[S_N]$:
  - 10
- $E[X]$:
  - $\frac{1}{2}$
- Can be shown that $E[N] < \infty$
- So, by Wald's equation, $E[N]$:
  - 20

Example 2 (contd.)

- Recall:
  - Let X_n, $n \in \{1,2,\dots\}$ be independent and s.t.:
    - $P(X_n = -1) = P(X_n = 1) = \frac{1}{2}$ for all $n \in \{1,2,\dots\}$
  - Let $N = \min\{n: X_1 + \cdots + X_n = 1\}$
  - N is a stopping time
- $E[S_N]$:
  - 1
- $E[X]$:
  - 0
- So, $E[S_N] \neq E[N]E[X]$ and Wald's equation is not applicable
- Hence, $E[N] = \infty$

Example 3 (contd.)

- Recall:
  - Consider a sequence of i.i.d. coin tosses with $P(\text{heads}) = p \in (0,1)$
  - For $n \in \{1,2,\dots\}$:
    - $X_n = \begin{cases} 1, & \text{if heads occurs on toss } n, \\ 0, & \text{else.} \end{cases}$
  - Let:
    - N be toss on which first heads occurs
    - $K = N - 1$ be number of tails prior to first heads
  - N is a stopping time, K is not
- $E[X]$:
  - p
- $E[S_N]$:
  - 1
- $E[N]$:
  - $\frac{1}{p}$, since N is geometric(p)
- Thus, $E[S_N] = E[N]E[X]$, which verifies Wald's equation
- $E[S_K]$:
  - 0
- $E[K]$:
  - $\frac{1-p}{p}$
- Thus, $E[S_K] \neq E[K]E[X]$, which shows that Wald's equation does not hold

A Corollary of Wald's Equation

- Let $X_1, X_2, \ldots$ denote the interarrival times of a renewal process
- Which one is the first renewal after time t?
  - ❑ renewal $(N(t)+1)$
- **Claim**: $(N(t)+1)$ is a stopping time for the sequence $X_1, X_2, \ldots$
- **Proof**:
  - ❑ $N(t)+1 = n \Leftrightarrow X_1 + \cdots + X_{n-1} \leq t, X_1 + \cdots + X_n > t$
  - ❑ Hence, the event $\{N(t)+1 = n\}$:
    - depends only on $X_1, \ldots, X_n$
    - is independent of $X_{n+1}, X_{n+2}, \ldots$
- **Corollary**: If $\mu < \infty$, then:
  - ❑ $E\left[S_{N(t)+1}\right] = \mu[m(t)+1]$
- **Proof**:
  - ❑ By Wald's equation, $E\left[\sum_{n=1}^{N(t)+1} X_n\right] = E[N(t)+1]E[X]$
- **Note**: $N(t)$ is *not* a stopping time for the sequence $X_1, X_2, \ldots$

limsup and liminf

- E.g.:
  - Let $s_n = \begin{cases} 1, & n \in \{1,3,\dots\}, \\ -1, & n \in \{2,4,\dots\} \end{cases}$
  - $\lim\limits_{n\to\infty} s_n$:
    - does not exist
- However, $\limsup\limits_{n\to\infty} s_n$ and $\liminf\limits_{n\to\infty} s_n$ exist and equal 1 and -1, respectively
- We say that the sequence $s_1, s_2, \dots$ has *limit superior* (or *upper limit*) U, denoted by:
  - $U = \limsup\limits_{n\to\infty} s_n$

  if:

  i. for every $\epsilon > 0$, there exists an integer N s.t. for all $n > N$, $s_n < U + \epsilon$, and

  ii. for every $\epsilon > 0$ and integer M, $s_n > U - \epsilon$ for some $n > M$
- These conditions either:
  - determine a finite U uniquely, or
  - no finite U exists, and we set $U = \infty$

limsup and liminf (contd.)

- Analogously, we define the *limit inferior* (or *lower limit*) L:
  - $L = \liminf_{n\to\infty} s_n$
- Alternatively, we can define it as:
  - $L = -\limsup_{n\to\infty}(-s_n)$
- **Exercise**: Show that (i) $U \geq L$ and (ii) $\lim_{n\to\infty} s_n = l$ iff $U = L = l$
- In applications, we often prove that $\lim_{n\to\infty} s_n = l$ by showing that:
  - $U \leq l$ and $L \geq l$
- **Note**: Here, we considered limits as $n \to \infty$, but the above concepts and results carry over in a straightforward manner to the case where limits are taken as $t \to \infty$

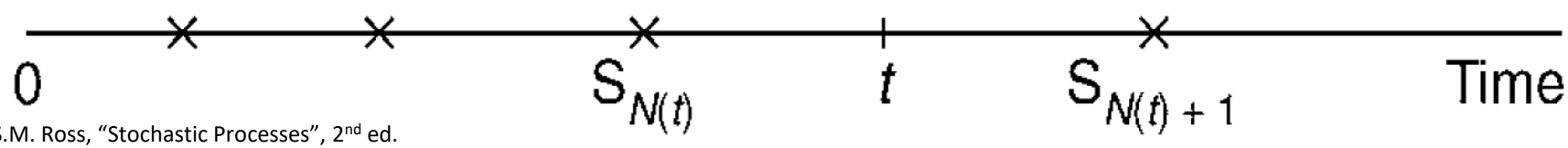
Proof of Elementary Renewal Theorem

- Recall:
  - **Theorem** (Elementary Renewal Theorem):

    2) $\frac{m(t)}{t} \to \frac{1}{\mu}$ as $t \to \infty$
  - **Corollary of Wald's Equation**: If $\mu < \infty$, then:
    - $E\left[S_{N(t)+1}\right] = \mu[m(t) + 1]$
- **Proof of Elementary Renewal Theorem**:
  - First, assume that $\mu < \infty$
  - We will prove the following, from which result follows:

    a) $\liminf_{t\to\infty} \frac{m(t)}{t} \geq \frac{1}{\mu}$

    b) $\limsup_{t\to\infty} \frac{m(t)}{t} \leq \frac{1}{\mu}$
  - $S_{N(t)+1} > t$; so, $\mu[m(t) + 1] > t$, from which a) follows
  - **Note**: Since $N(t)$ is not a stopping time for the sequence $X_1, X_2, \ldots$, we *cannot* prove b) by simply writing $\mu m(t) = E\left[S_{N(t)}\right] \leq t$ and taking limits
  - We use *truncation*: fix a constant $M > 0$ and consider the new renewal process with $\bar{X}_n = \min(X_n, M)$, for all $n \in \{1,2, \ldots\}$
  - $\bar{S}_{\bar{N}(t)+1} \leq t + M$; so, $\mu_M[\bar{m}(t) + 1] \leq t + M$
  - Hence, $\limsup_{t\to\infty} \frac{\bar{m}(t)}{t} \leq \frac{1}{\mu_M}$; since $\bar{m}(t) \geq m(t)$, we get $\limsup_{t\to\infty} \frac{m(t)}{t} \leq \frac{1}{\mu_M}$
  - Letting $M \to \infty$, we get b)
  - Result for case where $\mu = \infty$ can be similarly proved using truncated process and fact that $\mu_M \to \infty$ as $M \to \infty$



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Renewal Reward Processes

- Consider a renewal process $\{N(t), t \geq 0\}$ that has interarrival times $X_1, X_2, \ldots$ with CDF $F(\cdot)$
- Suppose each time a renewal occurs, we receive a reward
- Reward earned at time of nth renewal: R_n
- $R_1, R_2, \ldots$ are i.i.d. r.v.s
- For $n \geq 1$, R_n may depend on X_n
- Hence, the pairs (X_n, R_n), $n \geq 1$, are i.i.d.
- E.g.:
  - Consider a taxi that makes a series of trips
  - R_j: fare earned at end of trip j
  - X_j: time between collection of fares $(j-1)$ and j
  - Typically, R_j would depend on X_j (longer the trip, larger the fare)
- Let:
  - $R(t) = \sum_{n=1}^{N(t)} R_n$ be total reward earned until time t
  - $E(R) = E(R_n)$ and $E(X) = E(X_n)$

Renewal Reward Theorem

- **Theorem**: If $E(R) < \infty$ and $E(X) < \infty$, then:
  i. W.p. 1, $\frac{R(t)}{t} \to \frac{E(R)}{E(X)}$ as $t \to \infty$
  ii. $\frac{E[R(t)]}{t} \to \frac{E(R)}{E(X)}$ as $t \to \infty$
- **Proof**:
  - $\frac{R(t)}{t} = \left(\frac{\sum_{n=1}^{N(t)} R_n}{N(t)}\right)\left(\frac{N(t)}{t}\right) \to \frac{E(R)}{E(X)}$ as $t \to \infty$ by SLLN and strong law for renewal processes; this proves (i)
  - Since $N(t) + 1$ is a stopping time for $X_1, X_2, \ldots$, it is also a stopping time for $R_1, R_2, \ldots$
  - So, by Wald's equation, $E\left[\sum_{n=1}^{N(t)} R_n\right] = E\left[\sum_{n=1}^{N(t)+1} R_n\right] - E\left[R_{N(t)+1}\right] = (m(t) + 1)E(R) - E\left[R_{N(t)+1}\right]$
  - Hence, $\frac{E[R(t)]}{t} = \frac{(m(t)+1)}{t}E(R) - \frac{E\left[R_{N(t)+1}\right]}{t}$
  - Result follows from elementary renewal theorem if we show that:
    1) $\frac{E\left[R_{N(t)+1}\right]}{t} \to 0$ as $t \to \infty$
    - We omit the proof of 1)

Example

- A component may fail while in service, at cost c_f
- Alternatively, we can replace a working component that has been in service for time τ:
  - cost incurred is $c_r < c_f$
- Let V_j be in-service life of component and $V_1, V_2, \ldots$ be i.i.d. with CDF $G(\cdot)$
- Consider a policy, which replaces a component on failure or on reaching age T, whichever occurs first
- Want to find optimal value of T, i.e., value that minimizes the cost rate, say $k(T)$
- Time spent in service of component j:
  - $X_j = \min(V_j, T)$
- Note that $X_1, X_2, \ldots$ are i.i.d.
- Cost of jth replacement, say R_j:
  - $\begin{cases} c_r, & \text{if } V_j > T, \\ c_f, & \text{else.} \end{cases}$
- Cost rate $k(T)$:
  - $\frac{E(R_j)}{E(X_j)}$ by renewal reward theorem
- So, $k(T)$:

  1) $\frac{c_f G(T) + c_r G^c(T)}{\int_0^T G^c(x)dx}$
- Optimal value of T is one that minimizes quantity in 1)

Partial Rewards

- So far, we have assumed that reward R_n is earned all at once at end of nth renewal cycle, i.e., at time S_n
- However, renewal reward theorem also holds in more general case where:
  - reward is gradually earned during renewal cycle
- To show this, first suppose that all rewards are non-negative
- Then:
  - $\frac{\sum_{n=1}^{N(t)} R_n}{t} \leq \frac{R(t)}{t} \leq \frac{\sum_{n=1}^{N(t)} R_n}{t} + \frac{R_{N(t)+1}}{t}$
  - Take limits as $t \to \infty$ and use fact that $\frac{E\left[R_{N(t)+1}\right]}{t} \to 0$ as $t \to \infty$
- Similar argument holds when all rewards are non-positive
- General case follows by breaking up the returns into their positive and negative parts and applying above argument separately to each

Example

- Suppose travellers arrive at a train depot in accordance with a renewal process with mean interarrival time μ
- A train leaves whenever there are N travellers waiting in the depot
- Costs incurred:
  - depot incurs cost at the rate of nc dollars per unit time whenever there are n travellers waiting
  - additional cost of K each time a train is dispatched
- Want to find average cost per unit time incurred by depot
- Suppose cycle is completed whenever a train leaves:
  - then above is a renewal reward process
- $E[\text{length of cycle}]$:
  - $N\mu$
- Let X_n be time between nth and $(n+1)$st arrival in a cycle
- $E[\text{cost of cycle}]$:
  - $E[cX_1 + 2cX_2 + \cdots (N-1)cX_{N-1}] + K$
  - $\frac{c\mu N(N-1)}{2} + K$
- Average cost per unit time:
  - $\frac{c(N-1)}{2} + \frac{K}{N\mu}$ by renewal reward theorem

Alternating Renewal Processes

- Consider a system that can be in one of two states: *on* or *off*
- Initially, it is on and remains on for time Z_1, then goes off and remains off for time Y_1; then goes on for time Z_2, then off for time Y_2, and so on
- We assume that:
  - ❑ $Z_1, Z_2, \ldots$ are i.i.d. with mean $E[Z]$
  - ❑ $Y_1, Y_2, \ldots$ are i.i.d. with mean $E[Y]$
  - ❑ Z_n and Y_n may be dependent
- $\lim_{t\to\infty} \frac{\text{Amount of on time in } [0,t]}{t}$:
  - ❑ $\frac{E[Z]}{E[Z]+E[Y]}$ w.p. 1 by renewal reward theorem

Age and Excess of Renewal Process

- Consider a renewal process with interarrival times $X_1, X_2, \ldots$
- *Age* at time t:
  - ❑ time from last renewal until t
  - ❑ given by $A(t) = t - S_{N(t)}$
- *Excess* or *residual life* at time t:
  - ❑ time from t until next renewal
  - ❑ given by $Y(t) = S_{N(t)+1} - t$ (see Fig.)
- E.g.:
  - ❑ Suppose the renewal process is generated by putting a component in use and then replacing it upon failure
  - ❑ $A(t)$ is in-use age of component and $Y(t)$ its remaining life

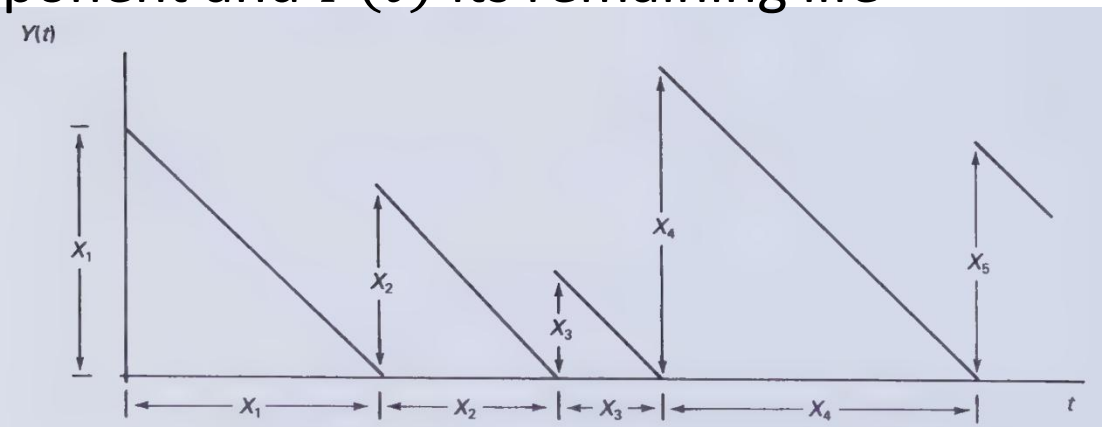


Figure 2-3

Ref: R. Wolff, "Stochastic Modeling and the Theory of Queues"

Age and Excess of Renewal Process (contd.)

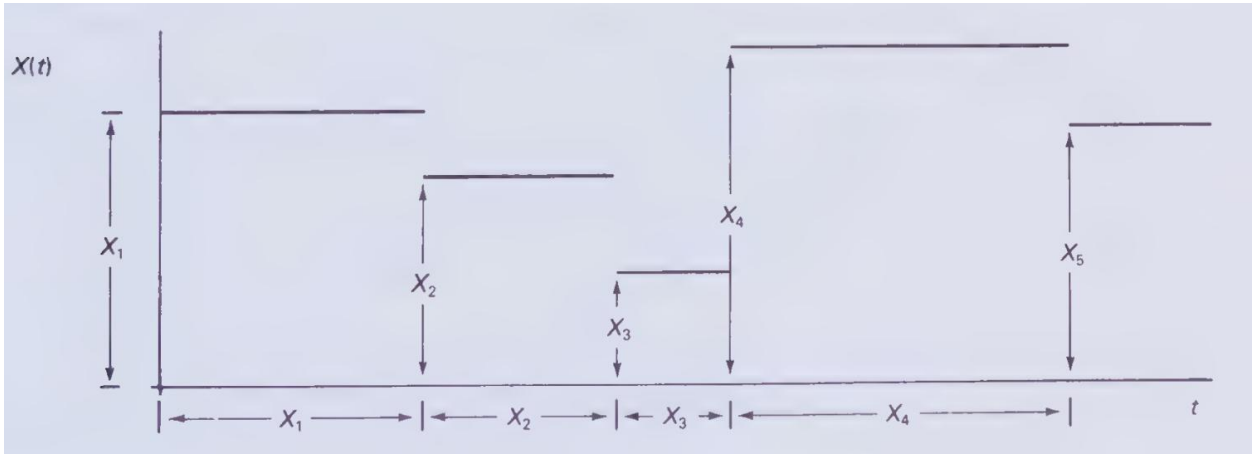
- Recall:
  - ❑ age at time t: $A(t) = t - S_{N(t)}$
  - ❑ excess at time t: $Y(t) = S_{N(t)+1} - t$
- Average age of process, $\lim_{t\to\infty} \frac{\int_0^t A(s)ds}{t}$:
  - ❑ $\frac{E\left(\int_0^X s ds\right)}{E(X)} = \frac{E(X^2)}{2E(X)}$ w.p. 1 by renewal reward theorem
- Similarly, average excess of process, $\lim_{t\to\infty} \frac{\int_0^t Y(s)ds}{t}$:
  - ❑ $\frac{E\left(\int_0^X (X-s)ds\right)}{E(X)} = \frac{E(X^2)}{2E(X)}$ w.p. 1 by renewal reward theorem
- Thus, average values of age and excess are equal
- Intuition:
  - ❑ Suppose the renewal process has been in operation for a long time
  - ❑ Then if we look backwards in time, the time between successive events will still be i.i.d. with the same distribution as the original process
  - ❑ Hence, looking backwards, we will see an identically distributed renewal process
  - ❑ But looking backwards, the excess life at t is exactly the age at t of the original process

Age and Excess of Renewal Process (contd.)

- Recall:
  - ❑ average age of process, $\lim_{t\to\infty} \frac{\int_0^t A(s)ds}{t} = \frac{E(X^2)}{2E(X)}$ w.p. 1
  - ❑ average excess of process, $\lim_{t\to\infty} \frac{\int_0^t Y(s)ds}{t} = \frac{E(X^2)}{2E(X)}$ w.p. 1
- Length of renewal interval containing the point t:
  - ❑ $X_{N(t)+1} = S_{N(t)+1} - S_{N(t)} = A(t) + Y(t)$
  - ❑ called "*spread*" of process
- Its average value:
  - ❑ $\lim_{t\to\infty} \frac{\int_0^t X_{N(s)+1}ds}{t} = \frac{E(X^2)}{E(X)}$ w.p. 1
- However, $\frac{E(X^2)}{E(X)} \geq E(X)$, with equality only when $\mathrm{var}(X) = 0$
- Thus, average value of $X_{N(t)+1}$ is $\geq E(X)$
- This is a consequence of the *inspection paradox* (details later)

Limiting Distribution of Spread

- Consider a renewal process with interarrival times $X_1, X_2, \ldots$
- Recall: "*spread*" of process (see Fig.):
  - ❑ $X_{N(t)+1} = S_{N(t)+1} - S_{N(t)} = A(t) + Y(t) = X_s(t)$ (say)
  - ❑ length of renewal interval containing the point t
- Want to find $\lim_{t\to\infty} \frac{1}{t}\int_0^t P[X_s(u) \le x]\, du$
- Suppose at time t, reward rate is $1_{\{X_s(t)\le x\}}$
- Reward obtained in renewal cycle j:
  - ❑ $R_j = \begin{cases} X_j, & \text{if } X_j \le x, \\ 0, & \text{else.} \end{cases}$
- $E(R_j)$:
  - ❑ $\int_0^\infty P(R_j > u)du$
  - ❑ $\int_0^x P(R_j > u)du$
  - ❑ $\int_0^x P(X_j \in (u, x])du$
  - ❑ $\int_0^x [F(x) - F(u)]du$
  - ❑ $xF(x) - \int_0^x F(u)du$

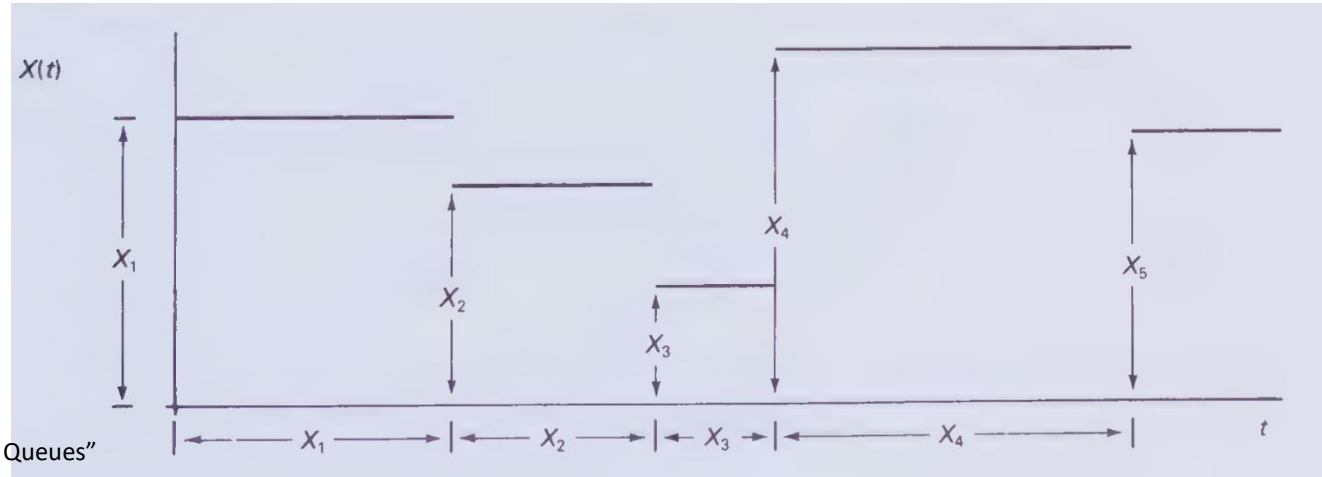


Ref: R. Wolff, "Stochastic Modeling and the Theory of Queues"

- Hence, $\lim_{t\to\infty} \frac{1}{t}\int_0^t P[X_s(u) \le x]\, du$:
  - ❑ $\frac{1}{\mu}\left[xF(x) - \int_0^x F(u)du\right] = F_s(x)$ (say) by renewal reward theorem

Limiting Distribution of Spread (contd.)

- Recall: $\lim_{t\to\infty} \frac{1}{t}\int_0^t P[X_s(u) \le x]\, du$:
  - ❑ $\frac{1}{\mu}\left[xF(x) - \int_0^x F(u)du\right] = F_s(x)$
- **Exercise**: Show that $F_s(x)$ is a valid CDF, i.e., it increases from 0 to 1 as x increases from $-\infty$ to ∞ and is right-continuous
- $F_s(x)$ is called limiting distribution of spread
- Corresponding PDF $f_s(x)$:
  - *1)* $\frac{xf(x)}{\mu}$, for $x \ge 0$
  - ❑ where $f(x)$ is the PDF corresponding to the CDF $F(x)$
- The factor x in 1) called "*length biasing*"
- Intuition:
  - ❑ a random observer is more likely to land in a long interval than a short one (there is more to land in)
  - ❑ this is reflected by biasing (weighing) the PDF at each x by interval length x
- **Exercise**: Show that the spread, $X_s(t)$, at any time t is stochastically larger than $X_j \sim F(\cdot)$. That is, show that for any $t > 0$:
  - ❑ $P(X_s(t) > x) \ge 1 - F(x)$, for all $x > 0$
- This is known as the *inspection paradox:*
  - ❑recall that we discussed its consequence earlier, viz., that w.p. 1, the average spread is $\frac{E(X^2)}{E(X)} \ge E(X)$



Ref: R. Wolff, "Stochastic Modeling and the Theory of Queues"

Batch Effects

- Discrete version of length biasing
- Suppose there are v_i children in family $i \in \{1,2,\dots\}$:
  - where $v_1, v_2, \dots$ are i.i.d. with PMF:
    - $P(v_i = j) = \alpha_j$, for $j \in \{0,1,\dots\}$
  - and mean:
    - $E(v_i) = E(v)$
- More generally, we refer to v_i as size of batch i
- Want to find:
  a) fraction of children that are members of batches of size $k \in \{1,2,\dots\}$
  b) fraction of children that are jth born, $j \in \{1,2,\dots\}$
  c) average order of birth of a randomly selected child
- To answer (b), let $v_i(j)$ be number of children in family i who are jth born
- Then $v_i(j)$:
  - $\begin{cases} 1, & \text{if } v_i \geq j, \\ 0, & \text{else.} \end{cases}$

Batch Effects (contd.)

- Fraction of children who are jth born:
  - $\lim_{n\to\infty} \frac{\sum_{i=1}^{n} v_i(j)}{\sum_{i=1}^{n} v_i}$
  - $\frac{\sum_{i\geq j} \alpha_i}{E(v)} = P(J = j)$ (say) a.s. by SLLN
  - where J denotes order of birth of a randomly selected child
- $E(J)$:
  - $\sum_{j=1}^{\infty} jP(J = j)$
  - $\frac{\sum_{j=1}^{\infty}\left(\sum_{i\geq j} j\alpha_i\right)}{E(v)}$
  - $\frac{\sum_{i=1}^{\infty} \alpha_i\left(\sum_{j=1}^{i} j\right)}{E(v)}$
  - $\frac{\sum_{i=1}^{\infty} \alpha_i\left(\frac{i(i+1)}{2}\right)}{E(v)}$
  - $\frac{E\{v(v+1)\}}{2E(v)}$
- Similarly, fraction of children that are members of batches of size k:
  - $\lim_{n\to\infty} \frac{\sum_{i=1}^{n} k1_{\{v_i=k\}}}{\sum_{i=1}^{n} v_i}$

  1) $\frac{k\alpha_k}{E(v)} = P(K = k)$,
  - where K is number of children in family of randomly selected child
- The factor k in 1) is called *batch size biasing*

Time Averages and Pointwise Limits

- Recall: in above discussion of limiting distribution of spread, we found:
  - $\lim_{t\to\infty} \frac{1}{t}\int_0^t P[X_s(u) \le x]\, du$
  - $\lim_{t\to\infty} \frac{1}{t}\int_0^t g(u)\, du,$
  - where $g(t) = P[X_s(t) \le x]$
- Such a limit is called a *time average* or *Cesaro limit (C.L.)* of g
- In contrast, consider a limit such as:
  - $\lim_{t\to\infty} P[X_s(t) \le x]$
  - $\lim_{t\to\infty} g(t)$
- Such a limit is called a *pointwise limit* of g
- How are these limits related?

Time Averages and Pointwise Limits (contd.)

- Recall:
  - time average/ *C.L.:* $\lim_{t\to\infty} \frac{1}{t}\int_0^t g(u)\,du$,
  - pointwise limit*:* $\lim_{t\to\infty} g(t)$
- Existence of time average does not imply that pointwise limit exists
- E.g.:
  - Let $g(t) = \begin{cases} 1, & \text{for } t \in (n-1, n], n \in \{1,3,5,\dots\}, \\ -1, & \text{for } t \in (n-1, n], n \in \{2,4,6,\dots\} \end{cases}$
  - Then $\lim_{t\to\infty} \frac{1}{t}\int_0^t g(u)\,du$:
    - 0
  - $\lim_{t\to\infty} g(t)$:
    - does not exist
- **Theorem**: If $\lim_{t\to\infty} g(t)$ exists and $\int_0^t g(u)\,du$ is finite for every finite t, then:
  - $\lim_{t\to\infty} \frac{1}{t}\int_0^t g(u)\,du = \lim_{t\to\infty} g(t)$
- Proof omitted

Example

- Consider the deterministic renewal process $X_j = 1, j \in \{1,2,\dots\}$
- For $y \in (0,1)$, $P(Y(t) \leq y)$:
  - $\begin{cases} 1, & \text{for } t \in [n-y, n), n \in \{1,2,\dots\}, \\ 0, & \text{else.} \end{cases}$
- $\lim_{t\to\infty} P(Y(t) \leq y)$:
  - does not exist
- $\lim_{t\to\infty} \frac{1}{t}\int_0^t P(Y(u) \leq y)\, du$:
  - y, for $y \in (0,1)$
- *Generalization*: Consider any renewal process in which $X_j, j \in \{1,2,\dots\}$, are integers
- Then $S_n, n \in \{1,2,\dots\}$, are also integers
- For $y \in (0,1)$, $\lim_{t\to\infty} P(Y(t) \leq y)$:
  - does not exist
- Even more generally, pointwise limits do not exist if $X_j, j \in \{1,2,\dots\}$, are *lattice r.v.s*

Lattice Random Variables

- A non-negative r.v. X is said to be *lattice* if $\exists d \geq 0$ such that $\sum_{n=0}^{\infty} P\{X = nd\} = 1$
- That is, X is lattice if:
  - it only takes on values that are integral multiples of some $d \geq 0$
- Largest d having this property called *period* of X
- If X is lattice and has CDF F, then we say that F is lattice

Examples

1) $P(X = 4) = P(X = 6) = \frac{1}{2}$:

- X is lattice with period 2

2) $P\left(X = \sqrt{2}\right) = P\left(X = \sqrt{8}\right) = \frac{1}{2}$:

- X is lattice with period $\sqrt{2}$

3) $P\left(X = \sqrt{2}\right) = P\left(X = \sqrt{3}\right) = \frac{1}{2}$:

- X is not lattice

Lattice Random Variables and Existence of Pointwise Limits

- Lattice versus non-lattice turns out to be the crucial distinction for:
  - existence versus non-existence of pointwise limits of many functions that arise in renewal theory
- Next, we discuss two theorems, which summarize these results:
  - Blackwell's Theorem
  - Key Renewal Theorem

Blackwell's Theorem

- **Theorem**:
  i. If F is not lattice, then for all $a \geq 0$:
     - $m(t+a) - m(t) \to \frac{a}{\mu}$ as $t \to \infty$

  ii. If F is lattice with period d, then:
     - $E[\text{number of renewals at } nd] \to \frac{d}{\mu}$ as $n \to \infty$
- We omit the proof, but provide intuition
- In the non-lattice case, suppose the limit $g(a) = \lim_{t\to\infty}[m(t+a) - m(t)]$ exists
- Then we show using the elementary renewal theorem that it must equal $\frac{a}{\mu}$
- $g(a+b)$:
  - ❑ $\lim_{t\to\infty}[m(t+a+b) - m(t)]$
  - ❑ $\lim_{t\to\infty}[m(t+a+b) - m(t+a) + m(t+a) - m(t)]$
  - ❑ $g(b) + g(a)$
- **Exercise**: Show that the only increasing solution of $g(a+b) = g(b) + g(a)$ with $g(0) = 0$ is $g(a) = ca$, for all $a > 0$, where c is a constant
- Let $x_n = m(n) - m(n-1)$ for $n \in \{1,2,\dots\}$
- Then $\lim_{n\to\infty} x_n = g(1) = c$ and hence:
  - ❑ $\lim_{n\to\infty} \frac{x_1 + \cdots x_n}{n} = c$
  - ❑ i.e., $\lim_{n\to\infty} \frac{m(n)}{n} = c$
- So, $c = \frac{1}{\mu}$ by elementary renewal theorem

Blackwell's Theorem (contd.)

- Recall:
  - **Theorem**:
    - i. If F is not lattice, then for all $a \geq 0$:
      - $m(t+a) - m(t) \to \frac{a}{\mu}$ as $t \to \infty$
    - ii. If F is lattice with period d, then:
      - $E[\text{number of renewals at } nd] \to \frac{d}{\mu}$ as $n \to \infty$

1) $g(a) = \lim_{t\to\infty}[m(t+a) - m(t)]$

- In the case where F is lattice with period d, the limit in 1) cannot exist:
  - since renewals occur only at points of the form nd
- If $\lim_{n\to\infty} E[\text{number of renewals at } nd]$ exists, then it can be shown using the elementary renewal theorem that it must equal $\frac{d}{\mu}$ (proof omitted)
- In the special case where interarrival times are always positive, part (ii) of Blackwell's theorem becomes:
  - $\lim_{n\to\infty} P[\text{renewal at } nd] = \frac{d}{\mu}$

Directly Riemann Integrable Functions

- Let h be a function defined on $[0, \infty)$
- For $a > 0$, let:
  - $\overline{m}_n(a)$ (respectively, $\underline{m}_n(a)$) be the supremum (respectively, infimum) of $h(t)$ over the interval $(n-1)a \leq t \leq na$
- h is *directly Riemann integrable* (d.R.i.) if $\sum_{n=1}^{\infty} \overline{m}_n(a)$ and $\sum_{n=1}^{\infty} \underline{m}_n(a)$ are finite for all $a > 0$ and:
  - $\lim_{a \to 0} a \sum_{n=1}^{\infty} \overline{m}_n(a) = \lim_{a \to 0} a \sum_{n=1}^{\infty} \underline{m}_n(a)$
- Differences between standard Riemann integrability (s.R.i.) and d.R.i.:
  - s.R.i is typically defined on a closed bounded interval $[a, b]$; to handle $[0, \infty)$, we take a limit as $b \to \infty$:
    - d.R.i. is directly defined on $[0, \infty)$
  - Under s.R.i., a function can be integrable even if it has wild oscillations as it heads towards infinity, as long as the positive and negative areas cancel out or settle down:
    - a d.R.i. function must be absolutely convergent, i.e., $\int_0^{\infty} |h(t)| dt < \infty$

Sufficient Condition

- The following is a sufficient condition for a function h to be d.R.i.:
  i. $h(t) \geq 0$ for all $t \geq 0$,
  ii. $h(t)$ is nonincreasing,
  iii. $\int_0^\infty h(t)dt < \infty$

Key Renewal Theorem

- Consider a renewal process with interarrival times $X_1, X_2, \ldots$ that have CDF F
- **Theorem**: If F is not lattice, and if $h(t)$ is d.R.i., then:
  - $\lim_{t\to\infty} \int_0^t h(t-x)dm(x) = \frac{1}{\mu}\int_0^\infty h(t)dt.$
- We omit the proof, but provide intuition
- By Blackwell's theorem:
  - $\lim_{t\to\infty} \frac{m(t+a)-m(t)}{a} = \frac{1}{\mu}$
- Hence:
  - $\lim_{a\to 0}\left(\lim_{t\to\infty} \frac{m(t+a)-m(t)}{a}\right) = \frac{1}{\mu}$
- Interchanging the limits, we get:
  - $\lim_{t\to\infty} \frac{dm(t)}{dt} = \frac{1}{\mu}$
- The key renewal theorem formalizes this
- Blackwell's theorem and the key renewal theorem can be shown to be equivalent
- Key renewal theorem is useful to compute the limiting value of $g(t)$, some probability or expectation at time t

CDF of $S_{N(t)}$

- Recall:
  1) For all $x \geq 0$, $m(x) = \sum_{n=1}^{\infty} F_n(x)$
- **Lemma**: For all t and s such that $0 \leq s \leq t$:
  - ❑ $P\{S_{N(t)} \leq s\} = F^c(t) + \int_0^s F^c(t-y)dm(y)$
- **Proof**: $P\{S_{N(t)} \leq s\}$:
  - ❑ $\sum_{n=0}^{\infty} P\{S_n \leq s, S_{n+1} > t\}$
  - ❑ $F^c(t) + \sum_{n=1}^{\infty} \int_0^{\infty} P\{S_n \leq s, S_{n+1} > t | S_n = y\} dF_n(y)$
  - ❑ $F^c(t) + \sum_{n=1}^{\infty} \int_0^{s} F^c(t-y) dF_n(y)$
  - ❑ $F^c(t) + \int_0^{s} F^c(t-y) d(\sum_{n=1}^{\infty} F_n(y))$

Limiting On Probability of Alternating Renewal Process

- Consider an alternating renewal process, where:
  - $Z_1, Z_2, \ldots$ are i.i.d. with CDF H and mean $E[Z]$
  - $Y_1, Y_2, \ldots$ are i.i.d. with CDF G and mean $E[Y]$
  - $Z_n + Y_n$ has CDF F
  - a renewal takes place each time the system goes on
- Recall: we showed that:

$$1)\ \lim_{t\to\infty} \frac{\text{Amount of on time in } [0,t]}{t} = \frac{E[Z]}{E[Z]+E[Y]} \text{ w.p. } 1$$

- This is a result about time average; next, we prove similar result about pointwise limit of:
  - $P(t) = P\{\text{system is on at time } t\}$

Limiting On Prob. of Alternating Renewal Process (contd.)

- **Theorem**: If $E[Z+Y] < \infty$ and F is non-lattice, then:
  - ❑ $\lim_{t\to\infty} P(t) = \frac{E[Z]}{E[Z]+E[Y]}$
- **Proof**:
  - ❑ Conditioning on time of last renewal prior to or at t:
    - ○ $P(t) = P\{\text{on at } t | S_{N(t)} = 0\} P\{S_{N(t)} = 0\} + \int_0^\infty P\{\text{on at } t | S_{N(t)} = y\} dF_{S_{N(t)}}(y)$
  - ❑ $P\{\text{on at } t | S_{N(t)} = 0\}$:
    - ○ $P\{Z_1 > t | Z_1 + Y_1 > t\} = \frac{H^c(t)}{F^c(t)}$
  - ❑ $P\{\text{on at } t | S_{N(t)} = y\}$:
    - ○ $P\{Z > t-y | Z+Y > t-y\} = \frac{H^c(t-y)}{F^c(t-y)}$, for $y \in (0,t)$
  - ❑ By above lemma, $dF_{S_{N(t)}}(y) = F^c(t-y)dm(y)$; hence:
    - ○ $P(t) = H^c(t) + \int_0^t H^c(t-y)dm(y)$
  - ❑ $H^c(t)$ is non-negative, non-increasing, and $\int_0^\infty H^c(t)dt = E(Z) < \infty$; also, $H^c(t) \to 0$ as $t \to \infty$:
    - ○ so, $H^c(t)$ is d.R.i.
  - ❑ Hence, by key renewal theorem, as $t \to \infty$:
    - ○ $P(t) \to \frac{\int_0^\infty H^c(t)dt}{\mu_F} = \frac{E[Z]}{E[Z]+E[Y]}$

Limiting Distributions of Age and Excess of Renewal Process

- Consider a renewal process with interarrival times $X_1, X_2, \ldots$ that have CDF F and mean μ
- Recall:
  - ❑ age at time t: $A(t) = t - S_{N(t)}$
  - ❑ excess at time t: $Y(t) = S_{N(t)+1} - t$
- **Theorem**: If F is non-lattice and $\mu < \infty$, then:
  - ❑ $\lim_{t\to\infty} P(Y(t) \le x) = \lim_{t\to\infty} P(A(t) \le x) = \frac{1}{\mu}\int_0^x F^c(y)dy$
- **Proof**:
  - ❑ Consider alternating renewal process in which system is on at time t iff $A(t) \le x$
  - ❑ By above theorem:
    - $\lim_{t\to\infty} P(A(t) \le x) = \frac{E[\min(X,x)]}{E(X)} = \frac{1}{\mu}\int_0^\infty P\{\min(X,x) > y\}dy = \frac{1}{\mu}\int_0^x F^c(y)dy$
  - ❑ Similarly, it can be shown that $\lim_{t\to\infty} P(Y(t) \le x) = \frac{1}{\mu}\int_0^x F^c(y)dy$

Limiting Distribution of Spread of Renewal Process

- Consider a renewal process with interarrival times $X_1, X_2, \ldots$ that have CDF F and mean μ
- Recall:
  - ❑ spread of process at time t: $X_s(t) = X_{N(t)+1} = S_{N(t)+1} - S_{N(t)} = A(t) + Y(t)$
- Earlier, we derived the time average of CDF of spread:

  1) $\lim_{t\to\infty} \frac{1}{t}\int_0^t P[X_s(u) \le x]\, du = \frac{1}{\mu}\left[xF(x) - \int_0^x F(u)du\right]$
- Now, we show that limiting distribution of spread equals the same value
- **Theorem**: If F is non-lattice and $\mu < \infty$, then:
  - ❑ $\lim_{t\to\infty} P[X_s(t) \le x] = \frac{1}{\mu}\left[xF(x) - \int_0^x F(u)du\right]$
- **Proof**:
  - ❑ Consider alternating renewal process in which system is on at time t iff $X_s(t) \le x$; then:
    - ○ $Z_j = \begin{cases} X_j, & \text{if } X_j \le x, \\ 0, & \text{else.} \end{cases}$
  - ❑ Recall: while proving 1), we showed that $E(Z_j) = xF(x) - \int_0^x F(u)du$
  - ❑ Result follows from above theorem about limiting on probability of alternating renewal process

Example

- Customers arrive at store, which sells single type of commodity, according to renewal process:
  - interarrival times $X_1, X_2, \ldots$ have non-lattice CDF F
- Amounts desired by customers are i.i.d. with CDF G
- Store uses (s, S) *ordering policy*:
  - If inventory level after serving a customer is below s, then order placed to bring it up to S
  - Otherwise, no order is placed
- Order is assumed to be instantaneously filled
- If inventory level after serving a customer is x, then amount ordered is:
  - $\begin{cases} S - x, & \text{if } x < s, \\ 0, & \text{else.} \end{cases}$
- Denote inventory level at time t by $X(t)$:
  - assume that $X(0) = S$
- Want to find $\lim_{t \to \infty} P[X(t) \geq x]$
- System is alternating renewal process:
  - on whenever inventory level is at least x and off otherwise
- By above theorem, $\lim_{t \to \infty} P[X(t) \geq x]$:
  - $\dfrac{E[\text{amount of time the inventory} \geq x \text{ in a cycle}]}{E[\text{time of a cycle}]}$

Example (contd.)

- Recall: $\lim_{t\to\infty} P[X(t) \geq x]$:

  1) $\dfrac{E[\text{amount of time the inventory} \geq x \text{ in a cycle}]}{E[\text{time of a cycle}]}$

- Let:
  - ❑ $Y_1, Y_2, \ldots$ denote the successive customer demands in a renewal cycle
  - ❑ $N_x = \min\{n\colon\ Y_1 + \cdots + Y_n > S - x\}$
- Which customer in renewal cycle causes inventory level to fall below x?
  - ❑ N_xth customer
- Which customer ends the renewal cycle?
  - ❑ N_sth customer
- Hence, by 1), $\lim_{t\to\infty} P[X(t) \geq x]$:
  - ❑ $\dfrac{E\left[\sum_{i=1}^{N_x} X_i\right]}{E\left[\sum_{i=1}^{N_s} X_i\right]}$
  - ❑ $\dfrac{E[N_x]}{E[N_s]}$
  - ❑ $\dfrac{1+m_G(S-x)}{1+m_G(S-s)}$, for $s \leq x \leq S$, since $N_x - 1$ can be interpreted as number of renewals until $S - x$ of a renewal process with interarrival times $Y_1, Y_2, \ldots$
- $m_G(t)$:
  - ❑ $\sum_{n=1}^{\infty} G_n(t)$

Delayed Renewal Processes

- Also known as general renewal processes
- Generalization of "ordinary renewal processes", which we have been studying so far
- Let $\{X_n: n = 1,2, \dots\}$ be sequence of independent non-negative r.v.s with X_1 having CDF $G(\cdot)$ and X_n having CDF $F(\cdot)$, for each $n > 1$
- As before, let:
  - ❑ X_n: time between $(n-1)$st and nth events
  - ❑ $S_0 = 0$ and $S_n = \sum_{i=1}^{n} X_i$ for $n \in \{1,2, \dots\}$,
  - ❑ where S_n is time of nth event
- $N_D(t) = \max\{n:\ S_n \leq t\}$
- **Definition**: The stochastic process $N_D(t)$ is called a *general* or a *delayed renewal process*
- E.g.:
  - ❑ We may start observing an ordinary renewal process at some time $t > 0$
  - ❑ If a renewal does not occur at time t, then time until first observed renewal will not have same distribution as that of future interarrival distributions
- Results similar to those for ordinary renewal processes hold for delayed renewal processes, as we now explain

Delayed Renewal Processes (contd.)

- $P\{N_D(t) = n\}$:
  - $P(S_n \leq t) - P(S_{n+1} \leq t)$
  - $G * F_{n-1}(t) - G * F_n(t)$
- Let $m_D(t) = E[N_D(t)]$
- Then for all $t \geq 0$, $m_D(t)$:
  - $\sum_{n=1}^{\infty} G * F_{n-1}(t)$

Limit Theorems for Delayed Renewal Processes

- Let $\mu = \int_0^\infty x dF(x)$
- **Proposition**:
  i. W.p. 1, $\frac{N_D(t)}{t} \to \frac{1}{\mu}$ as $t \to \infty$
  ii. $\frac{m_D(t)}{t} \to \frac{1}{\mu}$ as $t \to \infty$
  iii. If F is not lattice, then:
     - $m_D(t+a) - m_D(t) \to \frac{a}{\mu}$ as $t \to \infty$
  iv. If F and G are lattice with period d, then:
     - $E[\text{number of renewals at } nd] \to \frac{d}{\mu}$ as $n \to \infty$
  v. If F is not lattice, $\mu < \infty$, and $h(t)$ is d.R.i., then:
     - $\lim_{t\to\infty} \int_0^t h(t-x) dm_D(x) = \frac{1}{\mu} \int_0^\infty h(t) dt.$
- Above results can be proved by using corresponding results for ordinary renewal processes (details omitted)

Renewal Reward Theorem for Delayed Renewal Processes

- Consider a renewal reward process as defined earlier for ordinary renewal processes, but with the following changes:
  - X_1 has CDF $G(\cdot)$ and $E(X_1) < \infty$, and
  - R_1 may have a distribution that is different from that of $R_2, R_3, \ldots$; $E(R_1) < \infty$
- Let $E(X_n) = E(X)$ and $E(R_n) = E(R)$ for $n \geq 2$
- Renewal reward theorem discussed earlier still holds with above changes and is as follows (proof omitted)
- **Theorem**: If $E(R) < \infty$ and $E(X) < \infty$, then:

  i. W.p. 1, $\frac{R(t)}{t} \to \frac{E(R)}{E(X)}$ as $t \to \infty$

  ii. $\frac{E[R(t)]}{t} \to \frac{E(R)}{E(X)}$ as $t \to \infty$

Regenerative Processes

- Consider a stochastic process $\{X(t), t \geq 0\}$ with state space $\{0,1,2, \dots\}$ having following property:
  - there exist times $S_1, S_2, \dots$ at which the process probabilistically restarts itself
- That is, w.p. 1, there exists $S_1 > 0$ s.t. the continuation of the process beyond S_1 is:
  - probabilistic replica of the whole process starting at 0
- This implies existence of further times $S_2, S_3, \dots$ having same property as S_1
- Such a stochastic process is known as a *regenerative process*
- It follows from above that:
  - $S_1, S_2, \dots$ constitute the event times of a renewal process
- We say that a cycle is completed each time a renewal occurs

Regenerative Processes (contd.)

- Number of cycles until time t:
  - $N(t) = \max\{n\colon S_n \leq t\}$
- Let the CDF of a cycle duration be F
- **Note**:
  - The above process $\{X(t), t \geq 0\}$ is an *ordinary* regenerative process
  - *Delayed* regenerative processes defined similar to our definition of delayed renewal processes
  - For simplicity, we prove the following theorems for ordinary regenerative processes; they can also be shown to hold for delayed regenerative processes

Example

- Consider a generic queueing model in which customers arrive in accordance with a renewal process
- Consider the stochastic process $\{X(t), t \geq 0\}$, where:
  - $X(t)$ is number of customers in system at time t
- For most such queueing models (e.g., G/G/c or G/G/c/K), the process $\{X(t), t \geq 0\}$:
  - is a regenerative process with cycles beginning each time an arrival finds the system empty
- If the first customer arrives at time $t = 0$, then it is an *ordinary* regenerative process
- Otherwise, it is a *delayed* regenerative process

Regenerative Processes (contd.)

- **Theorem**: If F is non-lattice and $E[S_1] < \infty$, then:
  - $P_j = \lim_{t\to\infty} P\{X(t) = j\} = \frac{E[\text{amount of time in state } j \text{ during a cycle}]}{E[\text{duration of a cycle}]}$
- **Proof**:
  - Let $P(t) = P\{X(t) = j\}$
  - Conditioning on $S_{N(t)}$ and using lemma about CDF of $S_{N(t)}$, we get $P(t)$:
    - $P\{X(t) = j | S_{N(t)} = 0\}F^c(t) + \int_0^t P\{X(t) = j | S_{N(t)} = s\}F^c(t-s)dm(s)$
  - We have:
    - $P\{X(t) = j | S_{N(t)} = 0\} = P\{X(t) = j | S_1 > t\}$
    - $P\{X(t) = j | S_{N(t)} = s\} = P\{X(t-s) = j | S_1 > t-s\}$
  - So, $P(t)$:
    - $P\{X(t) = j, S_1 > t\} + \int_0^t P\{X(t-s) = j, S_1 > t-s\}dm(s)$
  - Now, $h(t) = P\{X(t) = j, S_1 > t\}$ is d.R.i. (exercise)
  - So, by the key renewal theorem, $\lim_{t\to\infty} P(t)$:
    - $\frac{1}{E[S_1]}\int_0^\infty P\{X(t) = j, S_1 > t\}dt$
    - $\frac{E\left[\int_0^\infty 1_{\{X(t)=j, S_1>t\}}dt\right]}{E[S_1]}$

Regenerative Processes (contd.)

- **Theorem**: If $E[S_1] < \infty$, then w.p. 1:
  - $\lim_{t \to \infty} \frac{[\text{amount of time in state } j \text{ during } (0,t)]}{t} = \frac{E[\text{amount of time in state } j \text{ during a cycle}]}{E[\text{duration of a cycle}]}$
- **Proof**:
  - Result follows from renewal reward theorem by:
    - letting a reward to be earned at rate 1 whenever the process is in state j and at rate 0 otherwise